Ontological Sphere Mapping: A Geometric Method for Visualizing Relations Between Sets

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Abstract

We introduce a novel visualization technique for representing relations between sets using a continuous 3D embedding of *spheres in semantic state space*. Each set is represented as a sphere whose radius reflects set size, while the volumetric intersection of spheres encodes overlap between sets. This approach generalizes Venn and Euler diagrams, which are limited to three sets, and provides quantitative, scalable, and interpretable visualizations of high-dimensional overlaps. We demonstrate the method using psychiatric nosology (DSM-5 disorders), where semantic axes correspond to ontological domains (Mood–Anxiety, Psychosis, Intrusions), and show how disorder overlaps can be geometrically quantified. We argue that *Ontological Sphere Mapping* offers a new paradigm for set visualization across science, medicine, and social systems.

1 Introduction

The visualization of overlapping sets is a long-standing challenge in information visualization. Traditional approaches such as Venn and Euler diagrams are expressive for up to three sets but scale poorly. Other methods (UpSet plots, network-based overlap graphs, matrix visualizations) represent overlap numerically but lose geometric intuition.

We propose a new technique, *Ontological Sphere Mapping (OSM)*, in which sets are represented as spheres embedded in a continuous metric space. Sphere radius is proportional to set size, while the volume of pairwise intersections reflects the degree of overlap. By anchoring the embedding in *semantic axes*, the method enables both quantitative analysis and interpretable visualization.

2 Method

2.1 Representation of Sets

Given a collection of sets $\{S_1, S_2, \ldots, S_n\}$, we define:

- Sphere center coordinates (x_i, y_i, z_i) in $\mathbb{R}^3$, determined either by semantic dimensions (ontology-driven) or dimensionality reduction (MDS, PCA) of a similarity matrix.
- Sphere radius r_i proportional to $|S_i|$, i.e. $r_i = \alpha + \beta|S_i|$ for scaling constants α, β.
- Pairwise overlap measured by volumetric intersection $V(S_i, S_j)$, computable from analytic formulae for intersecting spheres.

2.2 Intersection Volume

For two spheres with radii r_1, r_2 and center distance d, the intersection volume is:

$$V = \begin{cases} 0, & d \geq r_1 + r_2 \\ \frac{4}{3}\pi \min(r_1, r_2)^3, & d \leq |r_1 - r_2| \\ \frac{\pi(r_1+r_2-d)^2\left(d^2+2d(r_1+r_2)-3(r_1-r_2)^2\right)}{12d}, & \text{otherwise.} \end{cases}$$

This enables calculation of both absolute overlap volume and normalized percentage overlap relative to the smaller set.

2.3 Axes of Embedding

Axes can be defined in two ways:

1. **Ontology-driven:** Using theoretically meaningful dimensions (e.g., psychiatric domains such as Mood, Psychosis, Intrusions).
2. **Data-driven:** Using similarity matrices (e.g., Jaccard overlap) embedded via MDS or PCA.

3 Comparison to Existing Methods

- **Venn / Euler diagrams:** Limited to 2–3 sets, lack scaling. OSM generalizes to n sets.
- **UpSet plots:** Quantitative but not geometric. OSM preserves both.
- **Network graphs:** Capture overlaps via edge weights, but lose volumetric intuition. OSM encodes overlap as geometric volume.

To our knowledge, no existing set visualization method uses volumetric intersections of spheres in semantically meaningful spaces, making OSM a novel contribution.

4 Case Study: Psychiatric Nosology

We applied OSM to visualize relations among DSM-5 psychiatric disorders: Major Depressive Disorder (MDD), Generalized Anxiety Disorder (GAD), Post-Traumatic Stress Disorder (PTSD), Mania, Schizophrenia, and Obsessive-Compulsive Disorder (OCD).

- **Axes:** Mood–Anxiety, Psychosis, Intrusions.
- **Sphere sizes:** Number of DSM-5 diagnostic domains.
- **Overlap:** Disorders with multiple shared domains (e.g., PTSD–Schizophrenia) exhibited large intersection volumes, visualizing clinical comorbidity.

This revealed three major spectra: (i) Mood–Anxiety, (ii) Psychosis, (iii) Intrusion, with PTSD as a hub overlapping all domains.

5 General Applications

OSM is generalizable to:

- Genomics: genes $\times$ pathways $\times$ phenotypes.
- Law: rights $\times$ statutes $\times$ jurisdictions.
- Sociology: ideologies $\times$ value systems $\times$ networks.
- Data science: overlapping feature sets, user groups, or clusters.

6 Conclusion

Ontological Sphere Mapping (OSM) provides a novel, scalable, and semantically interpretable way to visualize overlapping sets. By embedding sets as spheres in a continuous state space, OSM unifies geometry, ontology, and quantitative overlap analysis. We believe this technique opens new possibilities for exploratory visualization across scientific domains.

References

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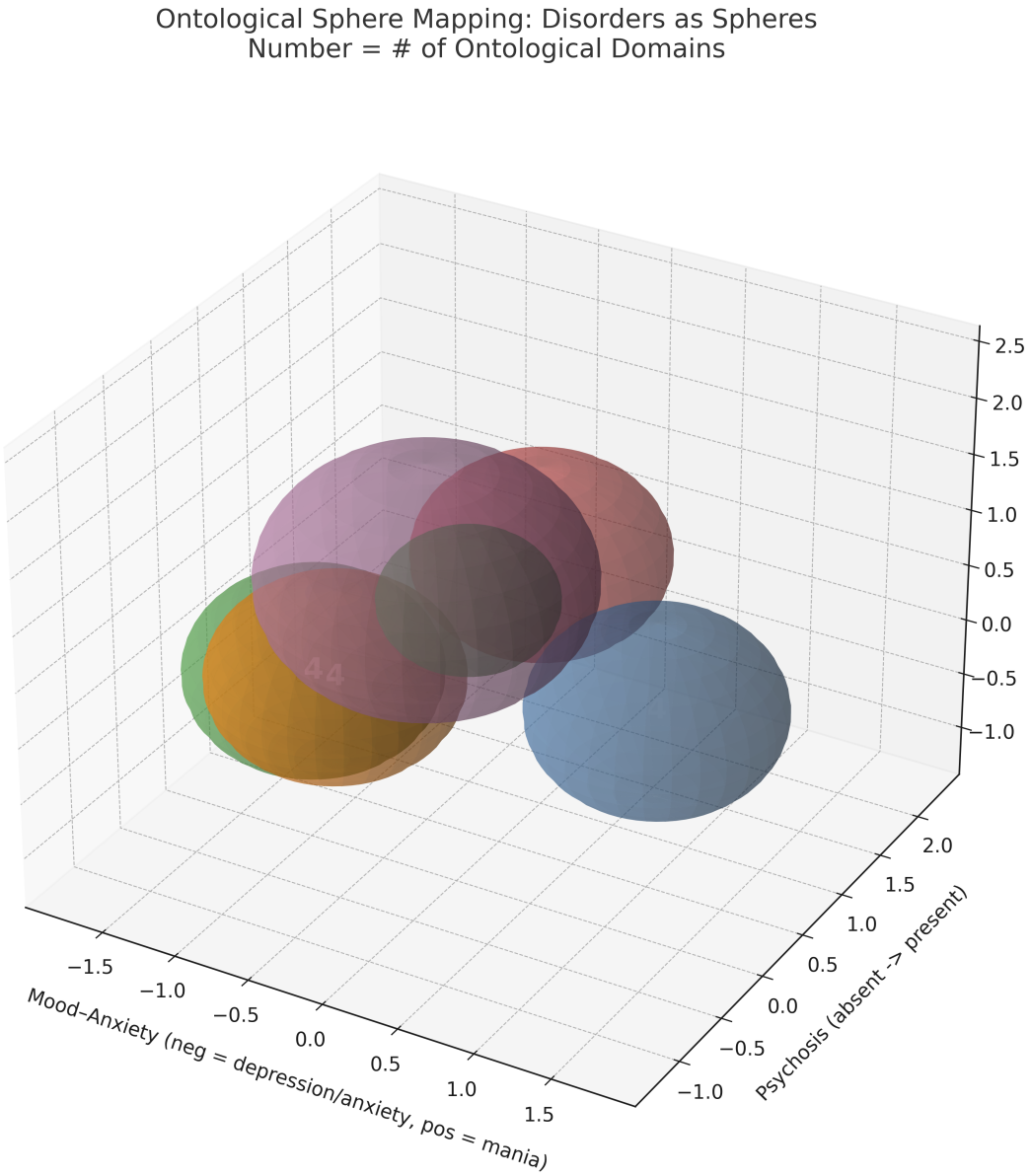

Figure 1: Ontological Sphere Mapping of DSM-5 disorders. Each disorder is represented as a sphere in a 3D state-space defined by Mood–Anxiety, Psychosis, and Intrusion axes. Sphere radius is proportional to the number of ontological domains spanned by the disorder. Numbers inside spheres indicate the exact domain count. Overlaps between spheres represent shared ontological domains, with volumetric intersections reflecting the magnitude of overlap.